EART27201— Sedimentary Rocks and Fossils
Fossil Practicals — Introduction and Corals

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Often there will be QR codes that link to online resources. If you are accessing the page via e.g. a laptop (so resolving the QR code is challenging), you can find a list of all the links at this URL:

https://www.russellgarwood.co.uk/EART27201.html

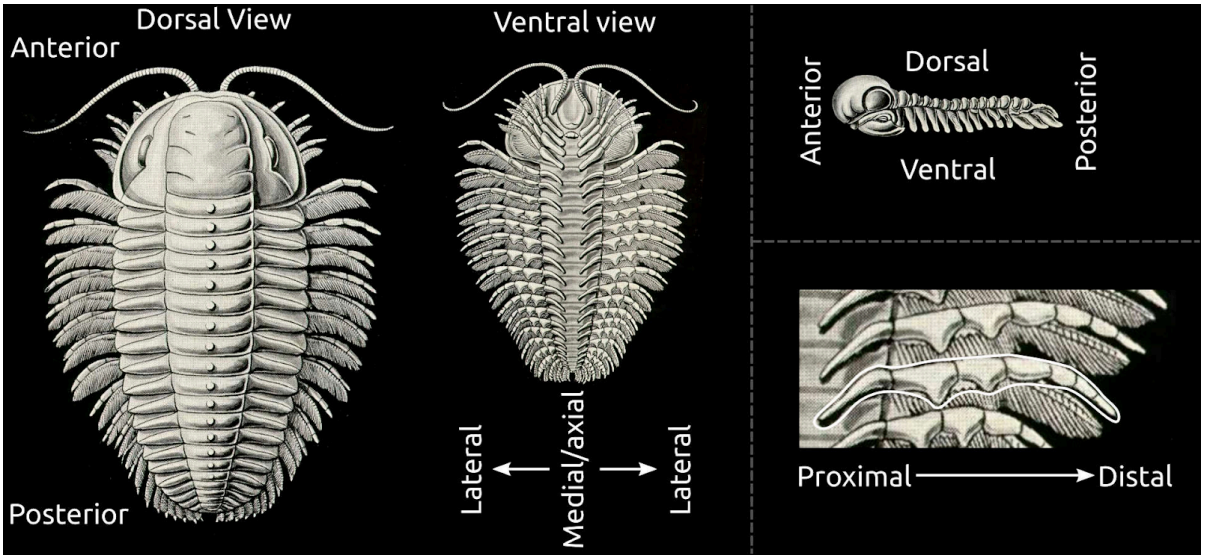
Under each QR code in this booklet is a keyword in italics - click the associated link on the page above, and it will take you to the correct target.

In this session we'll: cover some basics of biology and ecology; look at fossil preservation; and then we'll finish with a super fun exercise on corals. We'll mix lecture material and practicals: lectures will often give you the basics, and then the text in this document will expand upon those. Please do read the text before you try the exercises: it is there to provide context beyond what was in the lecture, and thus to help you.

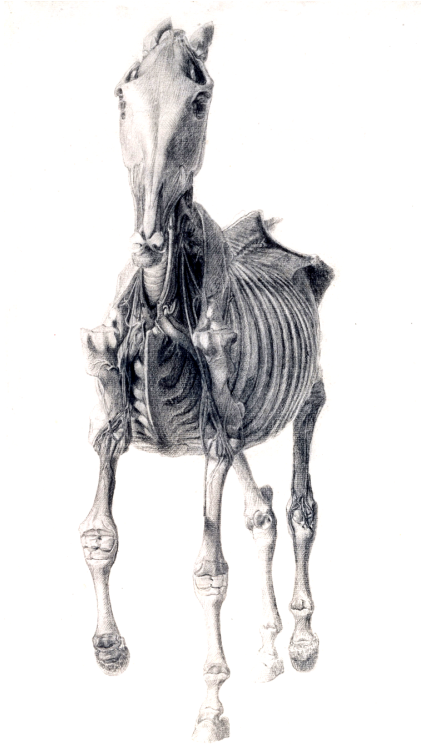
Let's start.

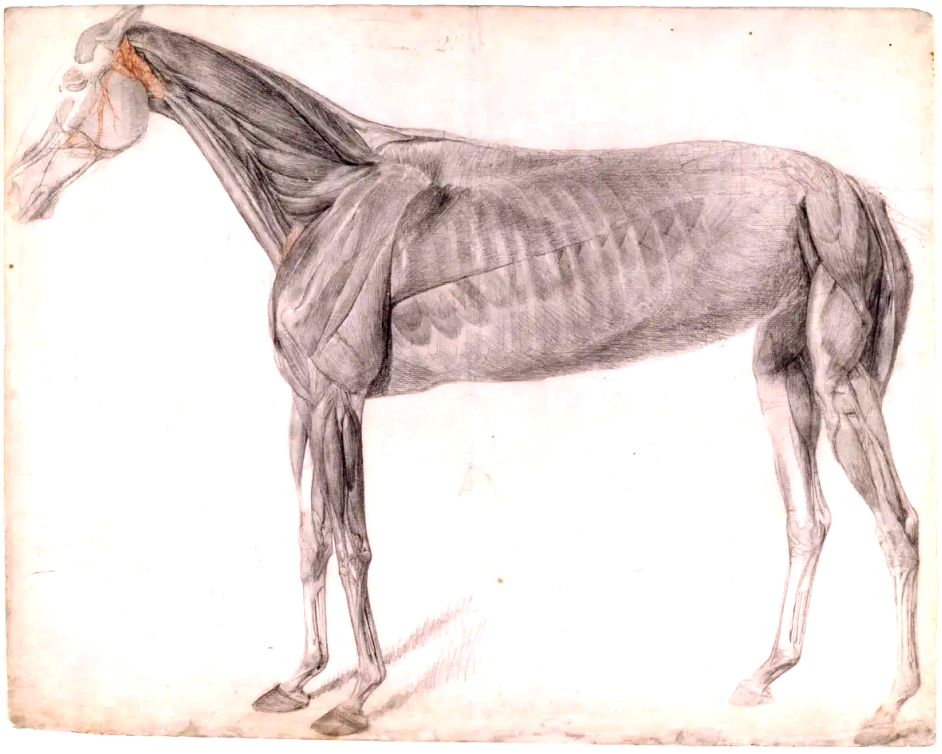
Anatomical terms of location

We have talked about how to communicate directions in animals through anatomical terms of location. Now let's put it into practice. Here is a recap, showing a trilobite (we will learn about these animals later in the course):



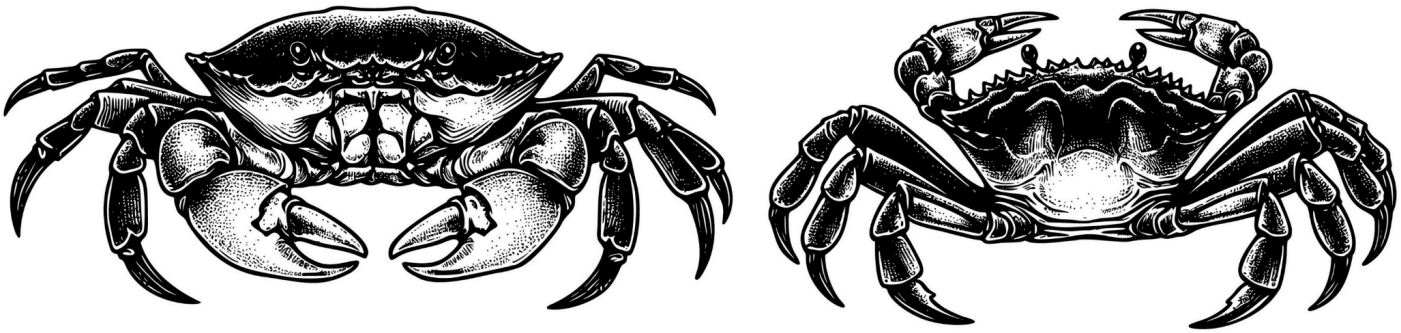
Now, let's see if you can label these on some other animals - let's start with a horse:



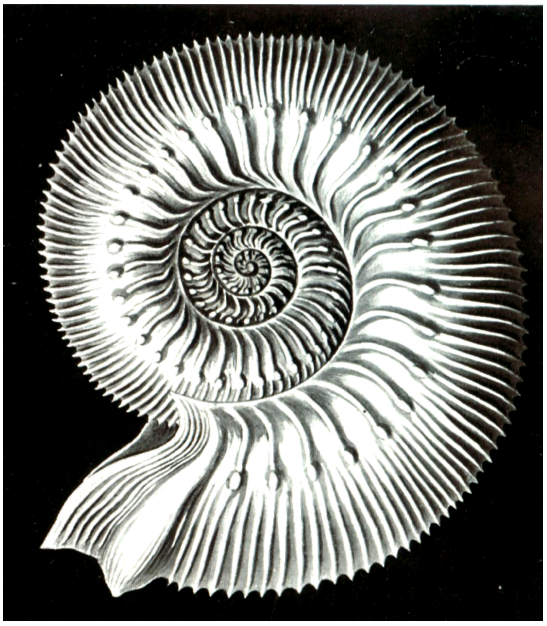


Please label dorsal, ventral, anterior, posterior, medial, and lateral on these amazing anatomical sketches of a horse (by a famous 18th century artist called George Stubbs, if you're interested…). On your favourite leg, please label the proximal and distal end.

Well done. Now let's do an invertebrate. Label the same on the picture of a crab below (image by Nur Jaeni):



Great. Now, because I'm really, really mean, what can you label on this? (It's a cephalopod, if you're wondering - we'll learn about them later in the course).



Please talk to Russell or one of the GTAs if you want to know what we can, and what we can't, say about these animals.

Ecology

We also met some terms describing the *ecology* of particular animals - i.e. where they live in a marine environment. These were:

Pelagic: Describing organisms that swim or drift in a sea or a lake, as distinct from those that live on the bottom (see benthos). Pelagic organisms are divided into plankton and nekton.

Benthic: Occurring on the bottom of a sea or lake. Benthic organisms may crawl, burrow, or remain attached to a substrate.

Plankton (Planktonic): The collection of pelagic organisms that drift or float passively with the current in a sea or lake. Plankton forms an important food source for many other members of the aquatic community and is divided into zooplankton and phytoplankton.

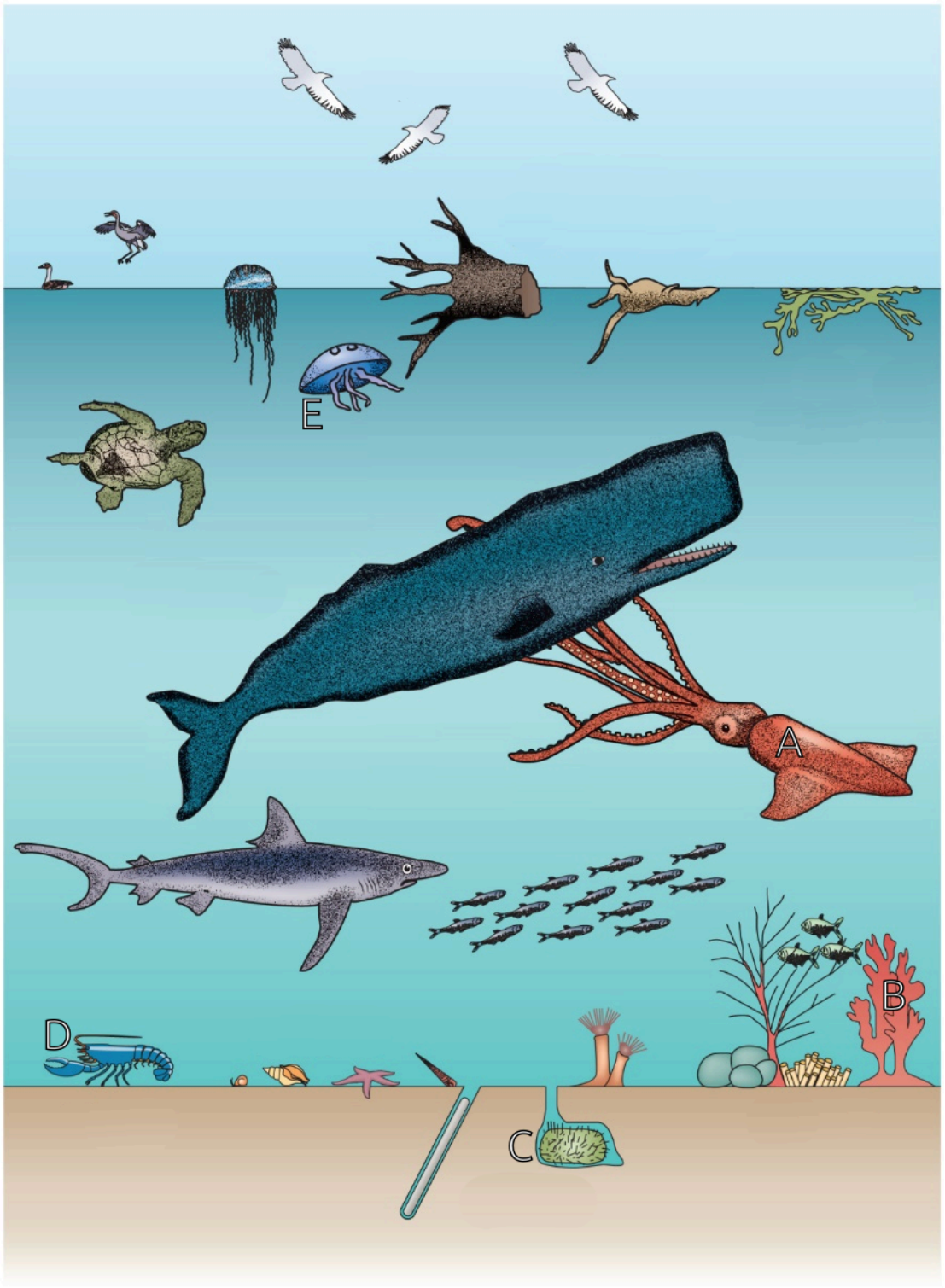
Nekton (Nektonic): Pelagic organisms that actively swim through the water.

Epifaunal: Applied to benthic organisms that live on the surface of the sea bed, either attached to objects on the bottom or free-moving. They are characteristic of the intertidal zone.

Infaunal: Applied to benthic organisms that dig into the sea bed or construct tubes or burrows. They are most common in the subtidal and deeper zones (i.e. the area seaward of the low-water mark).

Sessile: Describing animals that live permanently attached to a surface, i.e. sedentary animals. Many marine animals, e.g. sea anemones and limpets, are sessile.

Here is an illustration of a marine ecosystem, with some animals labelled A - E. For each, list as many of the above terms as you think apply to their mode of life.



Species A:

Species B:

Species C:

Species D:

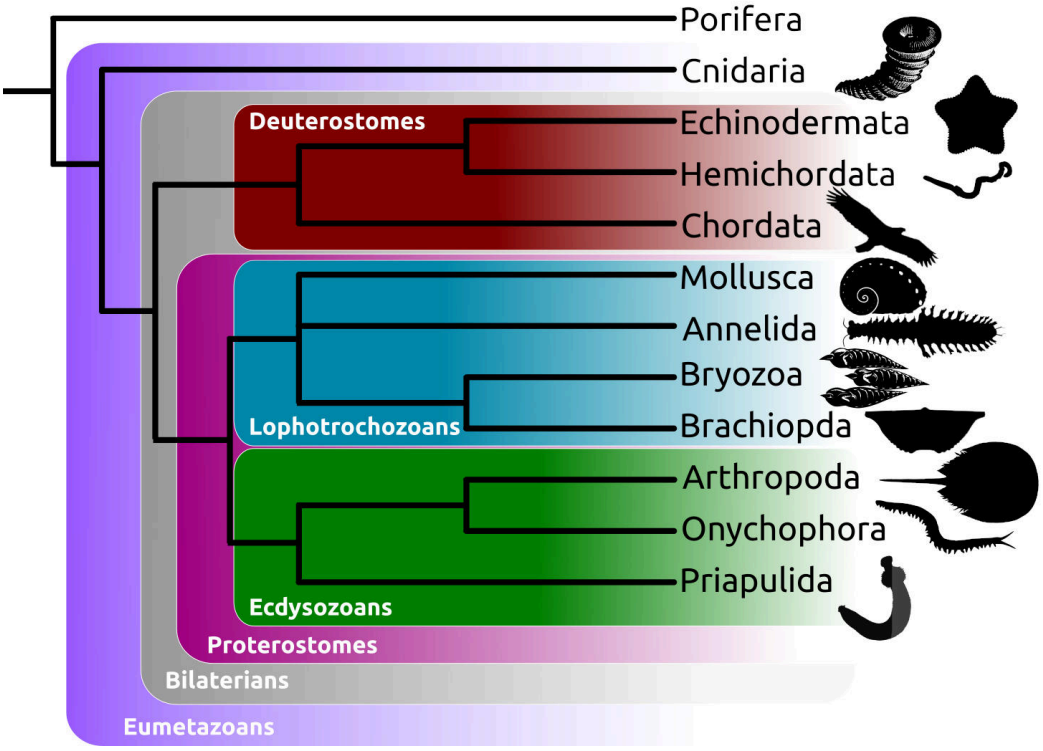
Species E:

If you want to check you are correct, please do ask a GTA, or Russell!

Phylogenies

Another important thing we covered was the tree of life - this is a really useful framework for learning about the living world. It places the groups we will learn about in a context that may well help you remember them. Below you will see that tree of life again (just so you are aware the relationships shown on this are far from settled - if you want to know more about how we build trees, and what it all means, the link on the right will take you to a website covering phylogenetics [this is not required for this course - it's bonus content in case you are interested. You are also always welcome to ask Russell questions. He loves phylogenetics]).

Link *Phylogenies*



So let's test our understanding of the terms we learned in the lecture. Note that all of the above terminals (= tips) on the phylogeny are at the *taxonomic rank* of Phylum.

What phyla form the sister group to the Priapulida?

The ecdysozoans and lophotrochozoans are sister clades. True or false?

What is the earliest branching branching animal phylum based on at this tree (i.e. it splits from other animals at the node nearest the root)?

The Annelida and Mollusca form a clade. Is this statement true or false?

The name for the Echinodermata + Hemichordata clade is the Ambulacraria. What is the sister group to the Ambulacraria?

Preservation

We talked briefly about the process of fossil preservation in the lecture. There are a number of different forms of preservation. Here's a quick overview.

Unaltered remains: Sometimes the hard parts of an organism are found in rocks unaltered. We can thus find the original materials (often biomineralised, but sometimes tissues) the organism created in life - shells are often a good example of this.

Permineralization: When the pore spaces in a porous material get filled with minerals this is called permineralization. This is common in - for example - bone and wood. Common minerals include calcite or silica, and are often sourced from groundwater. The original non-pore space may remain unaltered, or itself be replaced with minerals. A famous example of this form of preservation is found in the form of petrified wood, where the pore space is filled with minerals, and the wood is later replaced - thus no original wood is left.

Carbonised fossils: This kind of fossil occurs when organisms were buried quickly, and most often when there was little oxygen around. The remains of the organism are two-dimensional carbon films found on bedding planes. They often look black, because they are so carbon rich. The famous Burgess Shale, which you learned about last year, comprises carbonised fossils, but also this is relatively common in plant fossils, especially from the Carboniferous!

Replacement: Replacement happens when the original materials in a fossil are replaced with some form of secondary material - often shortly after burial. For example, in ammonites the mineral pyrite (FeS_2) can replace calcium carbonate to create a pyritized fossil.

External moulds: Sometimes a fossil can dissolve. The impression it leaves is a mould. Similarly, when you split a rock, on one part you will have the fossil, but you will find a mould on the other bit of rock you split off. If this mould represents the form of the outside, it is an external mould. These are relatively common within concretions - growths of a mineral around a fossil which harden, prior to its complete decay, leaving a void and a mould.

Internal moulds: Just sometimes, we get internal moulds. This occurs when, for example, an organism dies, and rots, but its shell was filled with sediment, or a mineral. If, then, the shell eventually dissolves - but what filled it does not - we are left with an internal mould! Sometimes, if you're lucky, you can find an external mould to go with the internal one.

Casts: These form when an external mould (or gap between an internal and external one) is filled with a substance that then lithifies. This reproduces the original form of the structure. So a mould is formed off an original structure, and a cast when a mould is filled.

You can see examples of each of these forms of preservation on the website associated with the introductory lecture:

Link *Preservation*

Fossil Preservation Task: Around the room you will find a range of different fossils, drawn from time periods throughout the Phanerozoic and across the tree of life. In the time available to us, try and look at a few of these and figure out which form of preservation each represents. Bear in mind that there are more than you can fit into the time we have available to us, so you don't need to do them all! This is primarily, as well as learning about fossil preservation, an opportunity to get a feel for handling fossils in person. It also serves as revision from last year, so hopefully some of these will feel familiar to you.

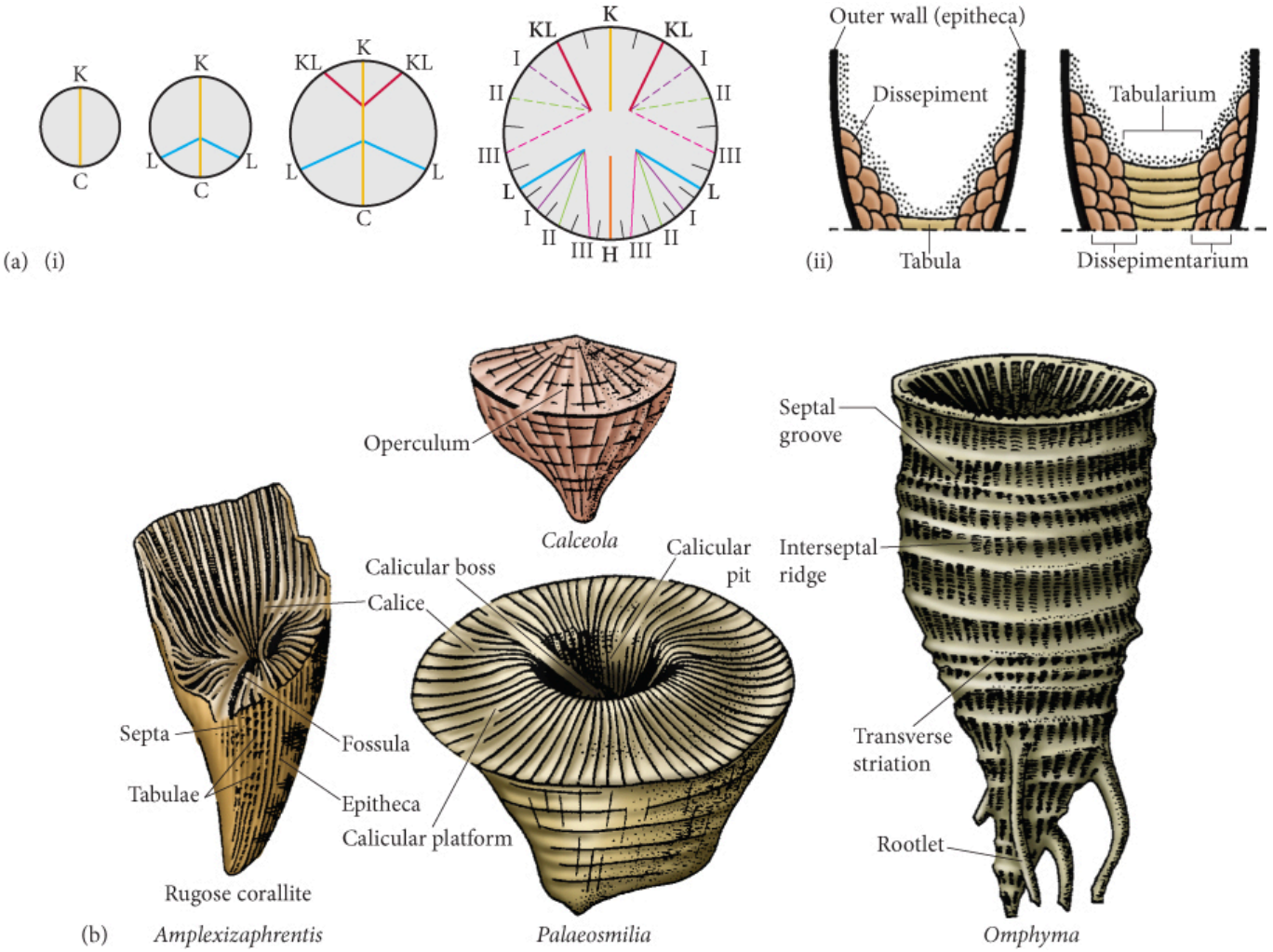
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Corals

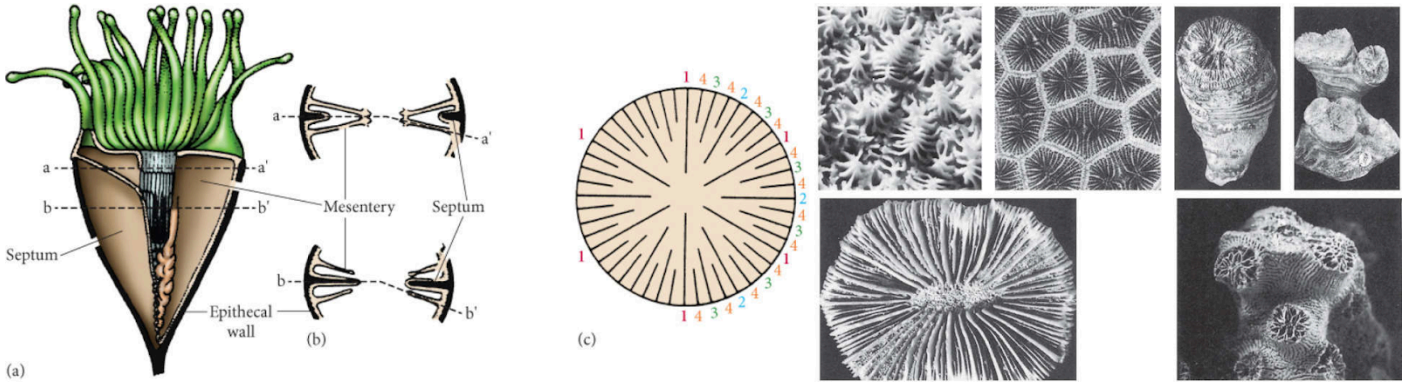
Orders

In our lecture we briefly met three different kinds of corals. Let's do so now in more detail.

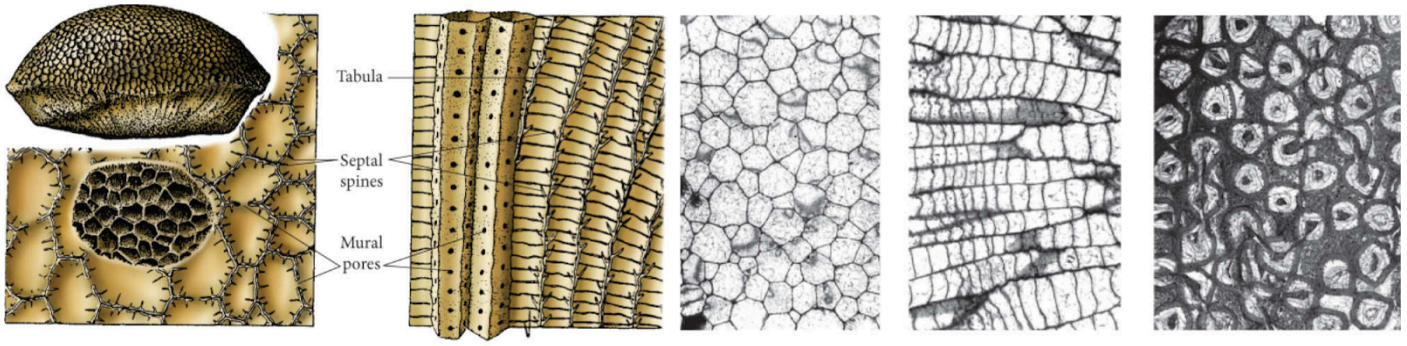
Rugosa (tetracorals): A Palaeozoic order of solitary and colonial corals which appears in the Ordovician and becomes extinct in the Permian. The corallum contains radial plates (septa) and horizontal plates (tabulae), and sometimes oblique plates (dissepiments). The septa developed in insertion cycles of four and many species retain a degree of bilateral symmetry. Solitary rugosans seem to have preferred soft substrates since they had no means of cementation and colonial forms seem to have relied for stability on the weight of the skeleton.



Scleractinia: Order of solitary or, more commonly, colonial corals, which always possess an external calcareous skeleton consisting essentially of radial partitions. Septa develop following the pattern of the radial infoldings in the body wall (mesenteries) in cycles of 6, 12, 24, 48, etc. The order first appeared in the Middle Triassic.



Tabulata: A Palaeozoic coral order comprising individuals that are always colonial, never solitary, and that possess small corallites. Tabulae are the major skeletal elements and septa are usually absent or poorly developed. The corallum is built of individual corallites, not always connected to each other, and may be encrusting, may consist of thin sheets, or may be massive. Other forms may be joined side by side to form fence-like structures. Smaller tabulates appear to have lived in deeper waters, whereas the larger tabulates appear in coral–stromatoporoid associations (stromatoporoidea are a group of sponges around from the Ordovician to Devonian).



If you want to see some 3D models for each group to get a clearer feeling for what each looks like before looking at some real specimens, you can find some 3D models here:

Link *Corals redux*

Corals Task 1: OK cool! Now let's see if you can identify these in hand specimen. You have access to two hand specimens. Have a close look at them, and using what you have learned from the lecture - and the definitions above - identify which coral group they belong to (these are technically different Orders - if you don't know what this means ask one of the GTAs or Russell). By the end of the session make sure you are able to tell apart members of these three orders. For each that you get a chance to look at, write a few notes justifying why you think it belongs to the order you place it in. When you are done with each, please place the fossils back in the correct box and put the fossils somewhere obvious so that others get the chance to look at them!

Siphonophyllia / Zaphrentis

Montlivaltia
Link *Emmonsia*	Emmonsia - scan the QR code to the left, or ask Russell to put it on the screen at the front.

Day/lunar month length

Presumably we're all happy with the concept that time passes. On Earth we tend to measure that relative to cycles. We have:

A (solar) year: The time it takes for Earth to revolve once around the sun, 365 days today.

A month: We often think of time in calendar months - each 1/12 of a year, as part of a calendar famously standardised by Pope Gregory XIII in 1582, building off that of Julius Caesar (46 BC). But there is also a lunar - or synodic - month. These reflect the natural orbital period of the Moon: there are 12.36 lunar months per year - this is why the full moon falls on different days of the month. It doesn't impact on our practical, but do bear in mind that there are also lunar calendars, where the seasons cycle through the months in a 30+ year cycle, and it's entirely arbitrary (and an artefact of history) that many of us are used to solar, rather than lunar calendars.

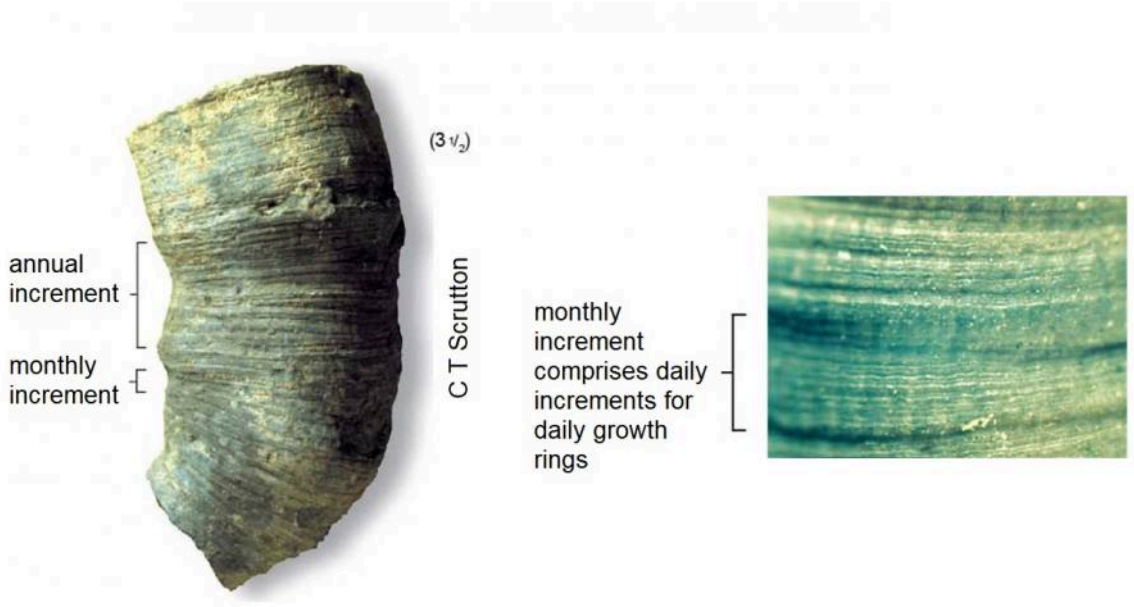
A day: The time it takes for the Earth to rotate once around its axis, today split into 24 hours.

Corals Task 2: *Do you think that these cycles have always been the same? If not, how have they changed?* Come up with a hypothesis for how these cycles may change with time from first principles that we can test, and then what it may mean for - as an example - day length. Feel free to talk to a GTA/RJG about this.

There are a number of ways we can address this question, but fossils - including corals - are one! How, I hear you ask? Well, let's find out by figuring out the length of a day in the Devonian period. Cool idea, huh?

In order for this to work, you need some key info:

- Today's corals grow a small amount every day. Each day's growth leaves a really small but visible growth line on the epitheca.
- Modern corals also have bands, comprising a number of growth lines. These seem to correspond with lunar cycles (our lunar - or synodic - month). These may be linked to tidal effects, nutrient variability, or a lunar breeding periodicity.
- Finally, some modern corals produced thickened growth bands (annulations), which are yearly. These might reflect changes in water temperature across seasons.



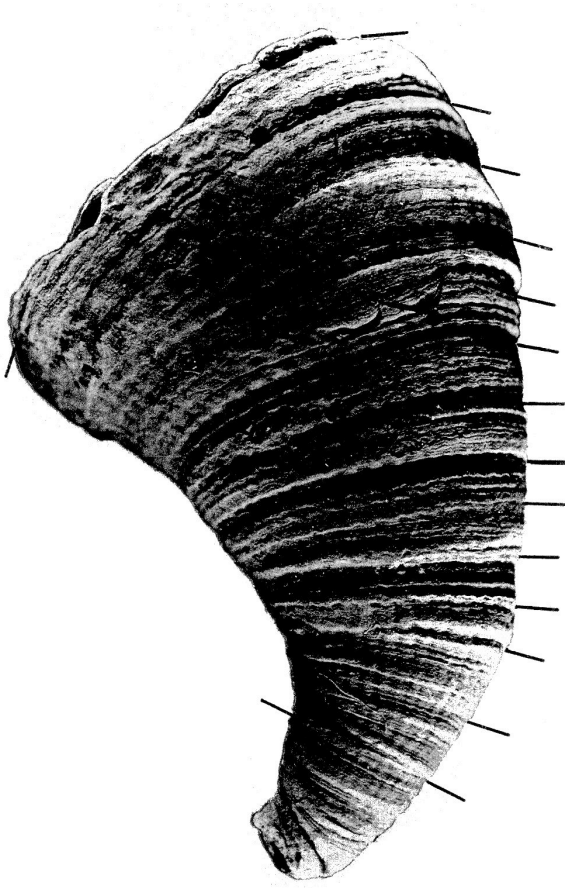
Heliophyllum halli (left) lived during mid Devonian times in Michigan, USA. Annual and monthly growth rings can be recognised. Source and copyright: BGS UK.

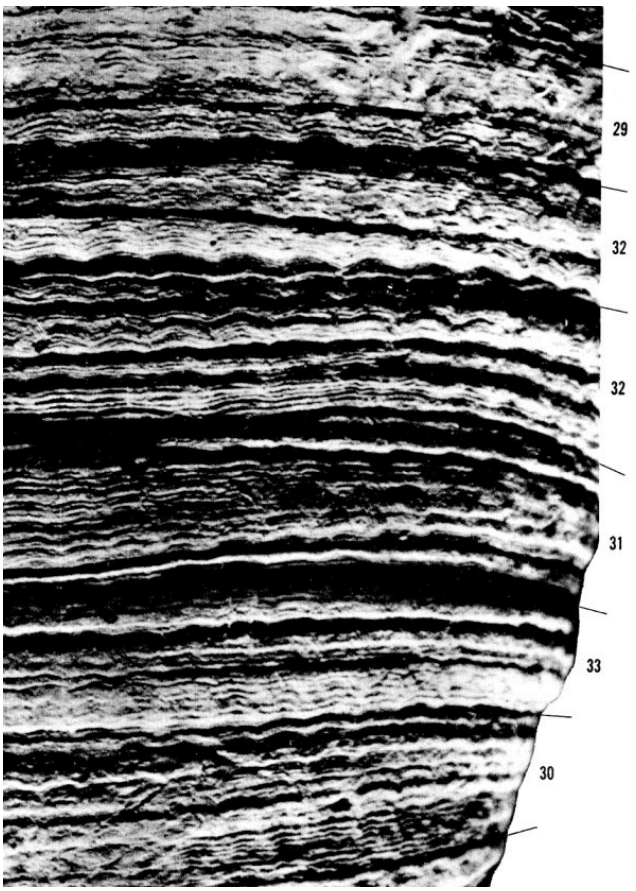
Corals Task 3: Let's assume (and this is an assumption) the same processes occurred and impacted on Palaeozoic coral growth. On the next page you will see an enlarged image of a Devonian Rugose coral, *Heliophyllum*. If you don't remember when the Devonian was, look it up. *Try and count the monthly bands on the right hand side of the image - I've left two marks on each side that the author considered possible annulations. I've also marked the first two bands on the right in grey to help you out. Based on your count, how many months do you think there may have been in the Devonian year?*


1
2

I recognise that this is something of an eye of faith situation, so here are the bands (below left) as identified by the author of the paper from which I took this example (Scrutton 1964 - reference at the end of this document). This author - who had access to a large number of specimens, and the benefit of experience (and of course, counting on the hand specimen), made 13.

Corals Task 4: On the right hand side image are counts from Scrutton (1964) of the daily growth lines for six of the monthly bands. *Calculate the average (you choose - mean, median, or mode) number of days per month in the Devonian from these counts. Note this down below, along with a justification of why in this circumstance the your chosen average (= measure of central tendency - mean, median, mode) is appropriate.*

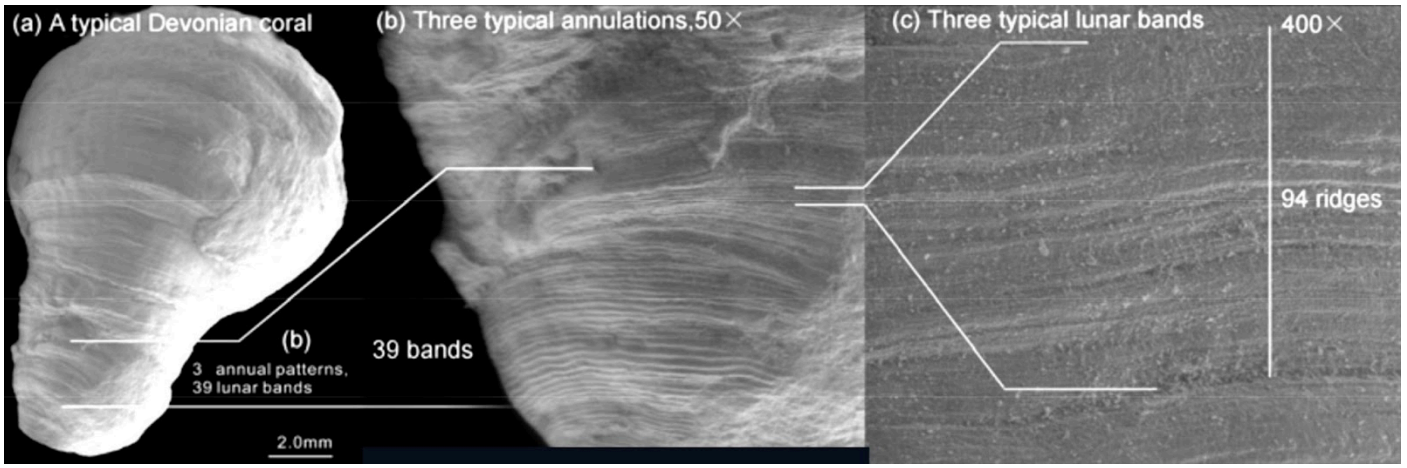




Corals Task 5: We now have a number of lunar months per (solar) year, and also an average number of days per month for the Devonian period. Cool huh? *Using this information calculate the number of days per solar year during the Devonian period.*

Great! Together we have worked out the number of days there were per year in the Devonian. As another example, here is an image from a more recent paper - this one based on a scanning electron microscope image - which has all the information required on it for you to work out the number of days per year using a different Devonian coral (~391.8 Ma; from Ahbach Fm, Mullertchen, Germany; source; Zhang et al 2010).

Corals task 6 (optional): *If you want to get another figure for the Devonian days per year, see if you can calculate it from the details below:*



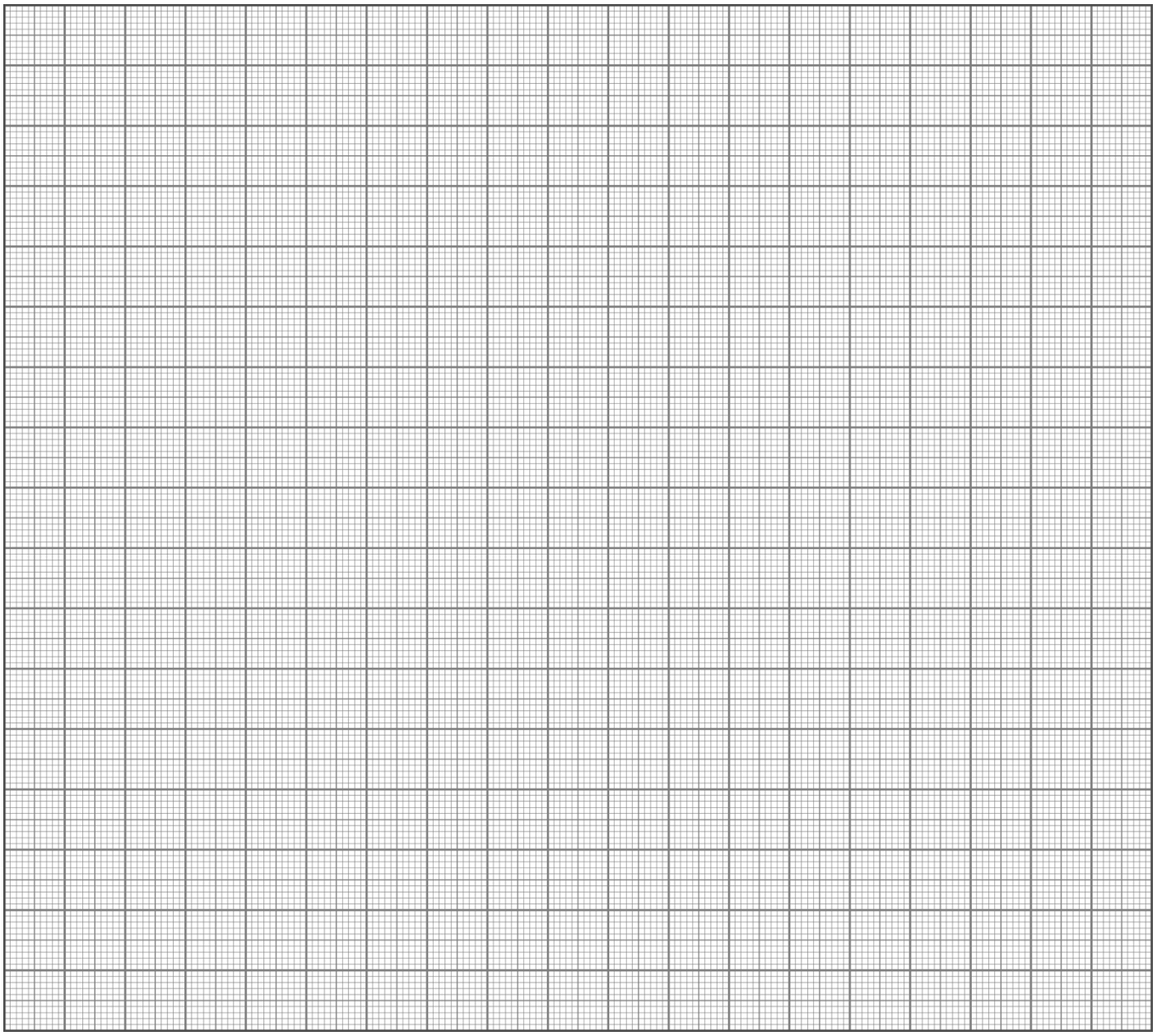
We could use the day length we have calculated and the day length where we are in the Holocene to say something about how day length has changed over time. But that would also be identifying a trend line on the basis of two points, which is not ideal. Thankfully, in the ~60 years since Scrutton (1964) was published, there have been a number more studies using a few different systems (i.e. different groups of organisms or other lines of evidence) to calculate the number of days per year. These are summarised in the table on the next page.

Area and formation	Geological period	Absolute age (Ma)	Average lunar months per year	Average days per year
Qiongzhusi Fm, Maotianshan, Chenjiang, China(Brachiopods)	Lower Cambrian, Terreneuvian	~535	13.3±0.1	436±5
Kaili Fm, Guizhou, China (Brachiopods)	Early Middle Cambrian	~525	13.2±0.1	435±8
Manger Fm, West Junggar Basin, China	Devonian, Pragian	~410	13.10±0.04	407±1
Bonar Village, Leon, Spain	Lower Devonian, Emsian	~405	13.1±0.1	405.8±0.5
Ahbach Fm-Nohn Fm, Eifel Mountains, Germany	Middle Devonian, Eifelian	391.8–397	13.07±0.07	406.3±0.5
Lower Hongguleleng Fm, China	Devonian, Frasinian	~375	13.05±0.05	406.2±0.5
Middle Hongguleleng Fm, China	Devonian, Famennian	~370	13.0±0.1	405.1±0.5
Upper Hongguleleng Fm, China	Devonian, Famennian	~365	13.0±0.1	404.2±0.5
Hebuke River Fm, China	Devonian, Famennian	~360	13.0±0.1	401.5±1.0
Liuzhou, Guangxi, China	Lower Carboniferous, Tournaisian	~350	13.0±0.1	399±5
Xibeikulasi Fm,West Junggar Basin, China	Lower Carboniferous, Visean	~340	13.03±0.05	394±5
Wuxuan, Guangxi, China	Lower Carboniferous, Visean	~335	13.0±0.1	392±5
Shiqiantan Fm, East Junggar Basin, China	Carboniferous, Moskovian	~310	12.98±0.05	392.5±0.5
Former USSR, from several formations in current Russian Federation	Carboniferous-Ordovician	260–440	(12.77–13.17)± 0.06, varies with age	(374–413)±1, varies with fossil age
Guanling Fm, Panxian, Guizhou, China (Bivalves)	Lower Triassic, Anisian	~240	12.70±0.05	373.0±2
Wayao Fm, Xinpu, Guanling, Guizhou, China (Bivalves)	Upper Triassic, Carnian	216–228	12.65±0.05	371.8±4
Treasured in Nanjing Institute of Geology and Paleontology, CAS	Cretaceous-Silurian	66–423	(12.45–13.14)± 0.03, varies with age	(370.0–403.6)±0.5, varies with fossil age
Tuchengzi Fm, Daohugou Beds, China (Bivalves)	Upper Jurassic, Kimmeridgian	~150	12.6±0.1	372.5±3
Yixian Fm, Liaoning, China	Lower Cretaceous, Aptian	~120	12.52±0.05	380.0±2
Caloosahatchee Fm, Florida, USA.	Quaternary, Pleistocene	~2	12.4±0.1	361.8±3

Corals task 7: *Using the data in this table and graph paper on the following page, construct a graph using appropriate axes of time against the average number of days per year. If you're feeling enthusiastic you can even include error bars from the data given.*

Note: You are welcome to do this using either spreadsheet software or using your R skills from the first year. You can find a csv file, which can be loaded in either, at the URL to the right.

Link *Coral data*



Corals task 8: Brilliant, well done for making it this far. Final thing to think about and then we're done - we've identified the trend, but what is its underlying cause. *Why is the Earth's rotation slowing? You may have already come up with a possible explanation when formulating your hypothesis for Task 2. If so, revisit it, and make sure you are happy that your results are in agreement with this hypothesis. If not, why not?*

Sources

Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press. (Definitions)

Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons. (Coral diagrams; marine ecosystem)

BGS. Corals. Discovering Geology — Fossils and geological time (Coral photo with growth rings)
https://www.bgs.ac.uk/discovering-geology/fossils-and-geological-time/corals/

Scrutton, C.T., 1964. Periodicity in Devonian coral growth. Palaeontology, 7(4), pp.552-558. (Monochrome images)

Zhang, W., Li, Z. and Lei, Y., 2010. Experimental measurement of growth patterns on fossil corals: Secular variation in ancient Earth-Sun distances. Chinese Science Bulletin, 55(35), pp.4010-4017.